

A projection-free formula for the non-inner Clark operator

Literature-identification note for arXiv:1205.4376

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Status. Literature-implied answer, complete for the selected open problem. Liaw–Treil arXiv:1308.3298 gives the requested formula without P_θ .

1 The question

Let θ be a scalar Schur function with $\theta(0) = 0$, put $\Delta = (1 - |\theta|^2)^{1/2}$, and let $\mathcal{K}_\theta$ be the two-component Sz.-Nagy–Foiaş model space. In Theorem 6.1 of [1], Liaw obtains, for $f \in C^1(\mathbb{T})$,

$$\Phi_1^* f(z) = P_\theta \left[f(z) + \int_{\mathbb{T}} \frac{\xi(f(\xi) - f(z))}{\xi - z} d\mu(\xi) \right] \begin{pmatrix} 1 - \theta(z) \\ -\Delta(z) \end{pmatrix}.$$

Remark 6.2(b), on PDF page 10, asks for an interpretation and, in particular, “a reasonable expression for $\Phi_1^* f$ that does not involve P_θ .” An image of the source page is preserved in the packet’s `figures` directory.

2 The later universal formula

Liaw and Treil treat the general Clark model in arXiv:1308.3298 [2]. Their Theorem 3.1, on PDF pages 11–12, applies to every transcription of the functional model and states

$$\Phi_\gamma^* f(z) = A_\gamma(z) f(z) + B_\gamma(z) \int_{\mathbb{T}} \frac{f(\xi) - f(z)}{1 - \bar{\xi}z} d\mu(\xi), \quad (1)$$

where $A_\gamma = c^\gamma$ and $B_\gamma = c^\gamma - z c_1^\gamma$. This is already a projection-free singular-integral formula. Equations (3.2)–(3.3) make the coefficient vectors explicit in the Sz.-Nagy–Foiaş transcription.

There is a harmless but important change of parameters. In the survey, $\mu = \mu_1$ is the spectral measure of the unitary Clark perturbation and $\Phi_1^* : L^2(\mu_1) \rightarrow \mathcal{K}_\theta$ models the contraction

$$U_0 = M_\xi - (\cdot, \bar{\xi}) 1.$$

In [2], the unperturbed unitary is named $U_1 = M_\xi$ on $L^2(\mu)$ and its contractive perturbation above is named U_0 ; its characteristic function is $\theta_0 = \theta$. Consequently

$$\Phi_1^* \text{ in [1] } = \Phi_0^* \text{ in [2].}$$

Setting $\gamma = 0$ in equations (3.1)–(3.3) of [2] gives

$$A_0(z) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad B_0(z) = \begin{pmatrix} 1 - \theta(z) \\ -\Delta(z) \end{pmatrix}.$$

Thus the requested formula is

$$\boxed{\Phi_1^* f(z) = \begin{pmatrix} f(z) \\ 0 \end{pmatrix} + \begin{pmatrix} 1 - \theta(z) \\ -\Delta(z) \end{pmatrix} \int_{\mathbb{T}} \frac{f(\xi) - f(z)}{1 - \bar{\xi}z} d\mu(\xi).} \quad (2)$$

It holds for $f \in C^1(\mathbb{T})$ under the assumptions of the source theorem and contains no P_θ . Moreover, (1) answers the same question for every $|\gamma| < 1$ and every model transcription.

3 Status and scope

The supporting paper cites the survey and develops its unpublished Liaw–Treil formula into the universal representation. Because identifying the source’s parameter 1 with the supporting paper’s parameter 0 requires the normalization above, this is recorded as a literature-implied answer. No claim is made about the survey’s unrelated spectral and localization questions.

References

- [1] C. Liaw, *Rank one and finite rank perturbations—survey and open problems*, arXiv:1205.4376.
- [2] C. Liaw and S. Treil, *Clark model in general situation*, J. Anal. Math. 130 (2016), 287–328; arXiv:1308.3298, Theorem 3.1.